

Project Euler 213

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1 Problem Breakdown

Firstly, upon breaking down the problem, we need to consider a start position (i), and find the probability that a flea in position i ends up at a certain position (S) in this 30x30.

The best way to do this is by simulating jumps in a 30x30 grid using transition probabilities in a Markov chain. To demonstrate this let's create a 5x5 grid and assume a flea begins at position (0, 0).

$$\begin{bmatrix} 1.0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We're representing this flea as a 1.0 since that is the only possible initial position of the flea. After one jump, the flea can move right or down, and since the chance is random and uniform, the new grid will look like this.

$$\begin{bmatrix} 0 & 0.5 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

So, after each jump we need to consider the neighbours of a square that could hold a flea and alter their probabilities accordingly. However, the problem requires us to find the expected number of **empty** squares. Therefore, we need to simulate all 900 possible start positions, creating a grid for each start position i that holds its probability distribution. We can then compute the following:

$$\sum_{S=1}^{900} \left[\prod_{i=1}^{900} (1 - P_i(S)) \right]$$

we find the product that all 900 fleas do not occupy a given square, and repeat this for all 900 squares.